

Algebraic dynamics on compact normal varieties

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Background & Motivation

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- (a) X is a normal $\mathbb{Q}$ -factorial projective variety with mild singularities $/k = \bar{k}$ with $\text{char } k = 0$.
- (b) $\text{Pic}(X)$, group of Cartier divisors of X modulo linear equivalence $\sim$.
- (c) $\text{NS}(X)$, Néron-Severi group of X .
- (d) $N^1(X) := \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$, $\rho(X) := \dim_{\mathbb{R}} N^1(X)$.
- (e) $\overline{\text{NE}}(X) := \overline{\{\sum a_i [C_i] \mid a_i \geq 0, C_i \text{ irreducible curve}\}}$.

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- If the canonical divisor K_X is nef, then we stop with the MM.
 - If K_X is not nef, then by the Cone Theorem, there is a morphism with connected fibres contracting a K_X -negative extremal ray $R \in \overline{\text{NE}}(X)$:

$$\pi : X \rightarrow Y$$

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onto a normal projective variety Y such that

- R is generated by some rational curve C .
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- Finally, we expect to stop the MMP with either a Fano contraction, or an MM (with K being nef).

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such that K_{X_r} is pseudo-effective, i.e., if we start from X with K_X being not p.e., then we can run a special MMP and end up with a Mori fibre space (Birkar-Cascini-Hacon-McKernan, 2010).

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2. Equivariant Minimal Model Program (EMMP)

Given any finite sequence of MMP from X , we say that it is **f -equivariant** if the following diagram commutes for some iteration $f_1 = f^s$:

$$\begin{array}{ccccccc} X = X_1 & \dashrightarrow & X_2 & \dashrightarrow & \cdots & \dashrightarrow & X_r \\ \downarrow f_1 & & \downarrow f_2 & & & & \downarrow f_r \\ X = X_1 & \dashrightarrow & X_2 & \dashrightarrow & \cdots & \dashrightarrow & X_r \end{array}$$

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Theorem (Existence of the EMMP)

EMMP holds in the following situations:

- *X is a Mori dream space (e.g. Fano varieties, toric varieties).*
- *(Nakayama) $\dim(X) = 2$ and $\deg(f) > 1$.*
- *(Meng-Zhang) X admits a polarized or int-amplified endomorphism.*

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Definition (Meng-Zhang, 2018; Meng, 2020)

Let $f : X \rightarrow X$ be a surjective endomorphism of a projective variety X .

- (i) f is **polarized**, if $f^*H \sim qH$ for some ample H and integer $q > 1 \Leftrightarrow f^*|_{N^1(X)}$ is diagonalizable with all eigenvalues being of modulus $q > 1$.

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- (iii) Clearly, “Polarized” $\Rightarrow$ “Int-amplified”

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Motivation and approach

- To classify compact Kähler spaces with dynamical restriction.
- (Campana-Das-Hacon-Höring-Peternell) The existence of the MMP and abundance in the Kähler setting have been confirmed in dim 3.
- (Claudon-Graf-Guenancia-Kirschner) The numerical characterization of Q -complex torus (a quasi-étale quotient of a complex torus) has been recently studied in the Kähler setting.

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1. First, we consider the EMMP for int-amplified endomorphisms.

Definition (Z., 2019)

Let $f : X \rightarrow X$ be a surjective endomorphism on a compact Kähler space with only rational singularities. Such f is **int-amplified**, if $f^*[\omega] - [\omega] = [\eta]$ for some Kähler classes $[\omega]$ and $[\eta] \Leftrightarrow$ all the eigenvalues of $f^*|_{H_{BC}^{1,1}(X)}$ on the Bott-Chern cohomology space are of modulus > 1 .

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Theorem A (EMMP for Kähler threefolds; Z., 2019)

Let $f : X \rightarrow X$ be an int-amplified endomorphism of a normal $\mathbb{Q}$ -factorial compact Kähler terminal threefold. Then, after iteration, there exist a $\mathbb{Q}$ -complex torus Y , a morphism $X \rightarrow Y$ and an f -EMMP of X over Y :

$$f \circ X = X_1 \xrightarrow{\pi_1} \cdots \xrightarrow{\pi_{i-1}} X_i \xrightarrow{\pi_i} \cdots \xrightarrow{\pi_{r-1}} X_r = Y \circ f_r$$

with each π_i a divisorial contraction, a flip or a Fano contraction, of a K_{X_i} -negative extremal ray, such that:

- If K_X is pseudo-effective, then $X = Y$ is a $\mathbb{Q}$ -complex torus.
- If K_X is not pseudo-effective, then each descended f_i (from f) on X_i is int-amplified and $X_i \rightarrow Y$ is an equi-dimensional holomorphic map with every fibre irreducible and rationally connected. Moreover, π_{r-1} is a Fano contraction.

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- (iii) Compared with the projective setting, we need to study the operation f^* on the Bott-Chern cohomology space and the Kähler cone instead of the Néron-Severi space and the ample cone.

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- (iv) In the non-projective setting, we possibly have no ample divisors.
- (v) Point (ii) can be further extended to any dimensional case due to a recent progress on the calculation of the **birational** second Chern class by Claudon, Graf and Guenancia.

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Let X be a $\mathbb{Q}$ -Gorenstein normal compact Kähler space with K_X being pseudo-effective. Suppose that X admits an int-amplified endomorphism. Then X is a $\mathbb{Q}$ -complex torus if X satisfies one of the following:

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Remark (Difficulties of dealing with general klt spaces)

We are not able to deal with the general klt case due to the bad behaviour of the birational second Chern class $c_2(X)$ and the second Chern class $c_2(\mathcal{T}_X)$ of the torsion-free sheaf $\mathcal{T}_X$.

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Problem (Dinh-Sibony, 2004; Dinh, 2012)

Classify compact Kähler manifolds X of dimension $n \geq 3$ admitting a free abelian group G of automorphisms of dynamical rank $n - 1$ (maximal) which is of positive entropy.

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Definition (Positive entropy, null entropy)

Let X be a normal projective variety (resp. a compact Kähler space with only rational singularities). We define the **first dynamical degree** $d_1(g)$ of an automorphism $g \in \text{Aut}(X)$ as the spectral radius of its natural pullback g^* on $N^1(X)$ (resp. $H_{\text{BC}}^{1,1}(X)$).

Such g is of **positive entropy** if $d_1(g) > 1$, otherwise it is of **null entropy**.

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- (iii) (Zhang, 2016) Let X be a normal projective variety as in the above problem. If X is not uniruled, then the G -equivariant minimal model program (G -MMP) exists and end up with a Q -abelian variety.

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- (iv) (Hu-Li, 2020) Hu and Li generalized and studied the above problem for projective varieties when G is of rank $n - 2$. (sub-maximal rank)
- (v) We generalized (iii) to the non-algebraic (singular) Kähler threefolds without running the G -MMP step by step.

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Theorem B (Characterization of $\mathbb{Q}$ -complex tori II; Z., 2021)

Let X be a $\mathbb{Q}$ -factorial compact Kähler terminal threefold. Suppose there is a free abelian automorphism group $G \cong \mathbb{Z}^r$ of positive entropy with $r = 2$ (maximal). Then we have:

- (1) If K_X is not pseudo-effective, X is rationally connected (projective).
- (2) If K_X is pseudo-effective, the Kodaira dimension $\kappa(X) = 0$ and X is bimeromorphic to $X_{\min} := T/H$ for a finite group H acting freely outside a finite set of a complex 3-torus T . Moreover, G descends to an automorphism subgroup $G|_{X_{\min}} \subseteq \text{Aut}(X_{\min})$ and further lifts to $G_T \subseteq \text{Aut}(T)$ such that $G|_{X_{\min}} \cong G_T/H$. (*G-equivariant*)

$$\begin{array}{c} T \curvearrowright G_T \\ \downarrow \\ G \curvearrowright X \dashrightarrow X_{\min} \curvearrowright G|_{X_{\min}} \end{array}$$

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The following is the key question in the second part of my thesis.

Question (Fakhruddin, 2002; Meng-Z., 2020)

Let X be a rationally connected (R.C.) smooth projective variety admitting a polarized or int-amplified endomorphism. Is X a toric variety?

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Definition (Toric varieties)

A normal projective variety X of dimension n is a **toric variety** if X contains an algebraic torus $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on X .

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The following result gives a partial answer to our key question.

Theorem (Yoshikawa, 2020)

A rationally connected (R.C.) smooth projective variety X is of Fano type if it admits an int-amplified endomorphism. (Toric varieties $\Rightarrow$ of Fano type)

II: Dynamical characterization of toric varieties

Theorem (Known results for our key question)

- (Nakayama, 2002) *A smooth rational surface is toric if and only if it admits a non-isomorphic surjective endomorphism. ✓ if $\dim X = 2$*
- (Amerik-Rovinsky-Van de Ven, 1999; Hwang-Mok, 2003) *A smooth Fano threefold of Picard number 1 is $\mathbb{P}^3$ if it admits a non-isomorphic surjective endomorphism. if ✓ $\dim X = 3$ and $\rho(X) = 1$*

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Theorem C (Characterization of toric varieties)

Let X be a R.C. smooth projective variety admitting an int-amplified f . Then X is toric if one of the following holds:

- (i) (Meng-Z., 2020) *f has totally invariant ramifications, i.e., there is a reduced divisor D on X such that $f|_{X \setminus D} : X \setminus D \rightarrow X \setminus D$ is étale.*
- (ii) (Meng-Zhang-Z., 2020) *X is a Fano threefold.*
- (iii) (Jia-Z., 2020) *X is a Fano fourfold admitting a conic bundle structure.*

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- For (iii), we generalize the idea developed in (ii) to the fourfold case. Here, it is important to choose a suitable factorization of the given (not necessarily elementary) conic bundle. (toric blow-ups)

III: (Totally) Invariant subvarieties on an algebraic variety

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- (3) Our next theorem is to give an optimal upper bound of the number of f^{-1} -invariant prime divisors in terms of $\dim(X)$ and $\rho(X)$. This generalizes Zhang's previous result on polarized endomorphisms.

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Theorem D (Boundedness of totally invariant divisors; Z., 2021)

(Simplified version) Let X be a $\mathbb{Q}$ -factorial klt projective variety of dim n , and f an int-amplified endomorphism on X . Let V_j ($1 \leq j \leq c$) be all the f^{-1} -invariant prime divisors. Then we have (with $\rho := \rho(X)$):

- (i) $c \leq n + \rho$. If $c \geq 1$, the pair $(X, \sum V_j)$ is lc and X is uniruled.
- (ii) Suppose $c \geq n + \rho - 2$. Then either X is R.C., or there is a fibration $X \rightarrow E$ onto an elliptic curve E such that every fibre is R.C.. In both cases, $(f^t)^*|_{\text{NS}_{\mathbb{Q}}(X)}$ is diagonalizable over $\mathbb{Q}$ for some t .
- (iii) Suppose $c \geq n + \rho - 1$. Then X is R.C..
- (iv) Suppose $c \geq n + \rho$. Then $c = n + \rho$, $K_X + \sum_{j=1}^{n+\rho} V_j \sim_{\mathbb{Q}} 0$ and f is étale outside $(\bigcup V_j) \cup f^{-1}(\text{Sing}(X))$. **the singular locus $\text{Sing}(X)$**

In particular, if c achieves the upper bound, then $(X, \sum V_j)$ is a toric pair, i.e., $\sum V_j$ is the complement of the big torus of X .

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- The motivation to study the finiteness of the maximal f -invariant SDD subvarieties comes from arithmetic geometry.
- Consider a dominant self morphism $f : X \rightarrow X$ of an algebraic variety. Let $V \subseteq X$ be a closed subvariety. We say that V is **f -periodic** (resp. **f -invariant**) if $\overline{f^s(V)} = V$ for some $s > 0$ (resp. $\overline{f(V)} = V$).

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- An f -periodic closed subvariety V is **SDD** if the 1st dynamical degree $d_1(f^s|_V) < d_1(f^s)$ for some $s > 0$ such that $\overline{f^s(V)} = V$.

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- Denote by Σ_{f^∞} the set of f -periodic SDD subvarieties. For $V \in \Sigma_{f^\infty}$, we say that V is **maximal** if $V \subseteq V'$ and $V' \in \Sigma_{f^\infty}$ imply $V = V'$.

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- Denote by S_{f^∞} the set of maximal elements in Σ_{f^∞} . Denote by S_f (resp. P_f) the subset of S_{f^∞} , consisting of all f -invariant elements (resp. prime divisors).

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Theorem E (Finiteness of maximal SDD; MMSZZ, 2020)

(E1). Let f be a dominant self-morphism of a variety X . Then

$$\#S_{f^n} \leq (d_1(f))^{n \cdot \dim(X)}(1 + o(1))$$

holds as $n \rightarrow \infty$, if (X, f) fits one of the following cases:

- (i) X is an algebraic group; and $f = R_a \circ g$ where g is an isogeny and R_a is a right multiplication by a (this is the case when X is semi-abelian).
- (ii) X is a toric variety with the big torus T being f -invariant.

(E2). Let f be a surjective endomorphism of a projective variety. Then the following holds as $n \rightarrow \infty$.

$$\#P_{f^n} \leq (d_1(f))^n(1 + o(1))$$

(E3). S_f is finite when X is either a projective surface or Q -abelian.

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Let X be a compact Kähler space with mild singularities. Is there an MMP for X ? Suppose that X admits an int-amplified endomorphism f . Can we run the MMP f -equivariantly? What about the end product?

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Thank you for your attention!

Q&A